浙江大学2016-17秋冬《数据结构基础》期末模拟练习

开始时间	1/1/2016, 12:00:00 AM	**结束时间**	1/18/2038, 12:00:00 AM	**答题时长**	120分钟
考生	–	**得分**	88	**总分**	100

判断题

总分：20　得分：14

1-1 For the following piece of code

```
if ( A > B ){
  for ( i=0; i<N*4; i++ )
    for ( j=N*N; j>i; j-- )
      C += A;
}
else {
  for ( i=0; i<N*N/20; i++ )
    for ( j=3*N; j>i; j-- )
      for ( k=0; k<N*3; k++)
         C += B;
}
```

the lowest upper bound of the time complexity is $O(N^4)$. (2分)

○ T　◉ F

评测结果：答案正确（2 分）

1-2 To sort N records, heap sort requires at least $O(N)$ extra space. (2分)

◉ T　○ F

评测结果：答案错误（0 分）

1-3 The best case time complexity of sorting algorithms based only on comparisons is at least $O(NlogN)$. (2分)

◉ T　○ F

评测结果：答案错误（0 分）

1-4 If `A` and `B` are both leaf nodes in a binary tree, then there exists a binary tree with preorder traversal sequence `...A...B...` and inorder traversal sequence `...B...A...`. (2分)

○ T　◉ F

评测结果：答案正确（2 分）

1-5 If a complete binary tree with 136 nodes is stored in an array (root at position 1), then the node at position 12 precedes the node at position 95 in its preorder traversal sequence. (2分)

○ T　◉ F

评测结果：答案正确（2 分）

1-6 If in a directed graph, every vertex is part of some cycle, then the graph must be strongly connected. (2分)

○ T　◉ F

评测结果：答案正确（2 分）

1-7 An algorithm to check for balancing symbols in an expression uses a queue to store the partial expression. (2分)

◉ T　○ F

评测结果：答案错误（0 分）

1–8 Given a hash table with size 13. If only the positions with odd (奇数) indices are occupied (the index starts from 0), then when the quadratic probing is used, insertion of a new key into this hash table can be successful. (2分)

◉ T ○ F

评测结果：答案正确（2 分）

1–9 Given two sorted lists $L1$ and $L2$, the fastest algorithm for computing $L1 \cup L2$ has time complexity $\Theta(NlogN)$. (2分)

○ T ◉ F

评测结果：答案正确（2 分）

1–10 Prim's algorithm is to maintain a forest and to merge two trees into one at each stage. (2分)

○ T ◉ F

评测结果：答案正确（2 分）

单选题

总分：60 得分：57

2–1 Let A and B be the two distinct nodes in a binary tree. A will be visited before B in the inorder traversal if __。（3分）

- ○ A. A is an ancestor of B
- ○ B. A is an descendant of B
- ○ C. A is on the right of B
- ◉ D. A is on the left of B

Note: Let R be the nearest common ancestor of A and B. "A is on the right (or left) of B" means that A is in the right (or left) subtree of R and B in the left (or right) subtree of R.

评测结果：答案正确（3 分）

2–2 Insert {5, 6, 7, 2, 4, 3} one by one into an initially empty binary search tree. The postorder traversal sequence of the resulting tree is _ (3分)

- ○ A. 2, 3, 4, 7, 6, 5
- ○ B. 2, 3, 4, 5, 6, 7
- ○ C. 5, 2, 4, 3, 6, 7
- ◉ D. 3, 4, 2, 7, 6, 5

评测结果：答案正确（3 分）

2–3 If the preorder and the postorder traversal sequences of a binary tree have exactly the opposite order, which of the followings is TRUE about the tree? (3分)

- ○ A. The tree is either empty or has only one node
- ◉ B. The tree height = number of nodes − 1 (Note: height is defined to be the number of edges along the longest path from the root.)
- ○ C. Any node in the tree has no left subtree
- ○ D. Any node in the tree has no right subtree

评测结果：答案正确（3 分）

2–4 Given an empty stack S and an empty queue Q. Push elements {1, 2, 3, 4, 5, 6, 7} one by one onto S. If each element that is popped from S is enqueued onto Q immediately, and if the dequeue sequence is {3, 2, 6, 5, 7, 4, 1}, then the minimum size of S must be: (3分)

- ○ A. 2
- ○ B. 3
- ◉ C. 4
- ○ D. 5

评测结果：答案正确（3 分）

2–5 Given a hash table of size 13 and the hash function h(x)=x mod 13. Assume that quadratic probing is used to solve collisions. After filling in the hash table one by one with input sequence {2, 15, 3, 16, 6, 25, 20, 7}, which number is placed in the position of index 7? (3分)

- A. 7
- B. 16
- C. 20
- D. none

评测结果：答案正确（3 分）

2–6 Which of the following statements is FALSE about the shortest path algorithms? (3分)

- A. Dijkstra's algorithm works out the shortest path in increasing order of distances.
- B. Single–source algorithms work fast on sparse graphs when finding the shortest path between any pair of vertices.
- C. Negative–cost edge in a directed graph will cause infinite loop in Dijkstra's algorithm.
- D. BFS can work out the shortest path when costs of all the edges are equal.

评测结果：答案正确（3 分）

2–7 The maximum flow in the network of the given Figure is: (4分)

- A. 104
- B. 110
- C. 132
- D. 125

评测结果：答案正确（4 分）

2–8 How many distinct topological sequences are there in the following DAG? (2分)

- A. 2
- B. 3
- C. 4
- D. 5

评测结果：答案正确（2 分）

2–9 Which of the following statements is TRUE about topological sorting? (3分)

- A. If a graph has a topological sequence, then its adjacency matrix must be triangular.
- B. If the adjacency matrix is triangular, then the corresponding directed graph must have a unique topological sequence.
- C. In a DAG, if for any pair of distinct vertices V_i and V_j, there is a path either from V_i to V_j or from V_j to V_i, then the DAG must have a unique topological sequence.
- D. If V_i precedes V_j in a topological sequence, then there must be a path from V_i to V_j.

评测结果：答案正确（3 分）

2–10 The array representation of the disjoint sets is given by {2, −4, 2, 3, −3, 5, 6, 9, −2}. Keep in mind that the elements are numbered from 1 to 9. After invoking Union(Find(4), Find(6)) with union–by–size, which elements will be changed in the resulting array? (3分)

- A. a. elements of index 2 and 5
- B. a. elements of index 2 and 6
- C. a. elements of index 4 and 5
- D. a. elements of index 4 and 6

评测结果：答案错误（0 分）

2–11 Given the adjacency list of a directed graph as shown by the figure. There is(are) __ strongly connected component(s). (3分)

- A. 4 {{0, 1, 5}, {2}, {3}, {4}}
- B. 3 {{2}, {4}, {0, 1, 3, 5}}
- C. 1 {0, 1, 2, 3, 4, 5}
- D. 1 {0, 5, 1, 3}

评测结果：答案正确（3 分）

2–12 Given input { 4321, 56, 57, 46, 289, 17, 331, 33, 234, 63 }. Which one of the following is the result after the 1st run of the Least Signification Digit (LSD) radix sort? (3分)

- A. →4321→331→33→63→234→56→46→57→17→289
- B. →331→4321→33→63→234→56→46→57→17→289
- C. →56→289→4321→331→33→234→46→57→63→17
- D. →57→46→289→17→33→234→63→56→4321→331

评测结果：答案正确（3 分）

2–13 Use Dijkstra's algorithm to find the shortest paths from vertex a to other vertices. In which order that the results must be obtained? (4分)

- A. 3 4 5 6 7
- B. 3 4 5 5 7
- C. 3 4 6 7 7
- D. 3 4 6 7 8

评测结果：答案正确（4 分）

2–14 To sort N records iteratively by merge sort, the number of runs is: (2分)

- A. $O(logN)$
- B. $O(N)$
- C. $O(NlogN)$
- D. $O(N^2)$

评测结果：答案正确（2 分）

2–15 If a DFS sequence of a graph is {V1, V4, V0, V3, V2}, then which one of the following can NOT possibly be the corresponding graph? (3分)

- A.
- B.
- C.
- D.

评测结果：答案正确（3 分）

2–16 Insert {10, 1, 14, 5, 8, 15, 3, 9, 7, 4, 11, 13} into an initially empty binary min–heap one at a time. After performing three DeleteMin operations, and then inserting 6 and 12 into the heap, the top element of the heap must be __ . (2分)

- A. 4
- B. 5
- C. 6
- D. 12

评测结果：答案正确（2 分）

2–17 Given a weighted graph as shown by the figure. Which one of the following statements is TRUE about its minimum spanning tree? (3分)

- (●) A. The minimum spanning tree is not unique and the total weight is 23
- ( ) B. The minimum spanning tree is unique and the total weight is 20
- ( ) C. The edge (B, F) must be in the tree and the total weight is 23
- ( ) D. The edge (H, G) must be in the tree and the total weight is 20

评测结果：答案正确（3 分）

2–18 Given the following four algorithms with their runtimes for problem size 100 and their time complexities:

Algorithm	Runtime	Time Complexity
A	100	$O(N)$
B	50	$O(N^2)$
C	20	$O(N^3)$
D	15	$O(N^4)$

Which algorithm is the fastest for problem size 200? (3分)

- ( ) A. A
- ( ) B. B
- (●) C. C
- ( ) D. D

评测结果：答案正确（3 分）

2–19 Insert 6 keys {23, 80, 95, 90, 82, 76} into an initially empty binary search tree. Suppose that the structure of the resulting tree is given in the figure. Which one of the followings is the possible sequence of insertions? (4分)

- ( ) A. 82 23 76 80 90 95
- ( ) B. 82 76 23 80 95 90
- (●) C. 82 90 76 23 95 80
- ( ) D. 82 90 95 80 76 23

评测结果：答案正确（4 分）

2–20 Given a binary tree with 100 leaves and without 1–degree nodes, the number of nodes in the tree is __ . (3分)

- ( ) A. 100
- ( ) B. 102
- (●) C. 199
- ( ) D. 200

评测结果：答案正确（3 分）

程序填空题

总分：12　得分：9

5–1 The function is to find the `K`–th largest element in a list `A` of `N` elements. The initial function call is `Qselect(A, K, 0, N-1)`. Please complete the following program.

```
ElementType Qselect( ElementType A[], int K, int Left, int Right )
{
    ElementType Pivot = A[Left];
    int L = Left, R = Right+1;

    while (1) {
        while ( A[++L] > Pivot ) ;
        while(A[--R]<Pivot)                    (3分);
        if ( L < R ) Swap( &A[L], &A[R] );
```

```
        else break;
    }
    Swap( &A[Left], &A[R] );
    if ( K < (L-Left) )
        return Qselect(A, K, Left, R-1);
    else if ( K > (L-Left) )
        return Qselect(A, K-(L-Left), R+1, Right)  (3分);
    else
        return Pivot;
}
```

评测结果：部分正确（3 分）

序号	结果	得分
0	答案正确	3
1	答案错误	0

5–2 The function is to find the `K`–th largest element in a list `A` of `N` elements. The function `BuildMinHeap(H, K)` is to arrange elements `H[1]` ... `H[K]` into a min–heap. Please complete the following program.

```
ElementType FindKthLargest ( int A[], int N, int K )
{   /* it is assumed that K<=N */
    ElementType *H;
    int i, next, child;

    H = (ElementType *)malloc((K+1)*sizeof(ElementType));
    for ( i=1; i<=K; i++ ) H[i] = A[i-1];
    BuildMinHeap(H, K);

    for ( next=K; next<N; next++ ) {
        H[0] = A[next];
        if ( H[0] > H[1] ) {
            for ( i=1; i*2<=K; i=child ) {
                child = i*2;
                if ( child!=K &&  H[child+1]<H[child]  (3分) ) child++;
                if (  H[0]>H[child]  (3分) )
                    H[i] = H[child];
                else break;
            }
            H[i] = H[0];
        }
    }
    return H[1];
}
```

评测结果：答案正确（6 分）

序号	结果	得分
0	答案正确	3
1	答案正确	3

函数题

总分：8　得分：8

6–1 LCA in BST

The lowest common ancestor (LCA) of two nodes `u` and `v` in a tree `T` is the deepest node that has both `u` and `v` as descendants. Given any two nodes in a binary search tree (BST), you are supposed to find their LCA.

Format of function:

```
int LCA( Tree T, int u, int v );
```

where `Tree` is defined as the following:

```
typedef struct TreeNode *Tree;
struct TreeNode {
    int   Key;
    Tree  Left;
    Tree  Right;
};
```

The function `LCA` is supposed to return the key value of the LCA of `u` and `v` in `T`. In case that either `u` or `v` is not found in `T`, return `ERROR` instead.

Sample program of judge:

```
#include <stdio.h>
#include <stdlib.h>

#define ERROR -1
typedef struct TreeNode *Tree;
struct TreeNode {
    int   Key;
    Tree  Left;
    Tree  Right;
};

Tree BuildTree(); /* details omitted */
int LCA( Tree T,  int u, int v );

int main()
{
    Tree T;
    int u, v, ans;

    T = BuildTree();
    scanf("%d %d", &u, &v);
    ans = LCA(T, u, v);
    if ( ans == ERROR ) printf("Wrong input\n");
    else printf("LCA = %d\n", ans);

    return 0;
}

/* Your function will be put here */
```

Sample Input 1 (for the tree shown in the Figure):

```
2 7
```

Sample Output 1:

```
LCA = 6
```

Sample Input 2 (for the same tree in Sample 1):

```
1 9
```

Sample Output 2:

```
Wrong input
```

代码

```
#ifdef PINTIA
#define ERROR -1
typedef struct TreeNode *Tree;
struct TreeNode {
    int   Key;
    Tree  Left;
    Tree  Right;
};
```

```
Tree BuildTree(); /* details omitted */
int LCA( Tree T,  int u, int v );

int main()
{
    Tree T;
    int u, v, ans;

    T = BuildTree();
    scanf("%d %d", &u, &v);
    ans = LCA(T, u, v);
    if ( ans == ERROR ) printf("Wrong input\n");
    else printf("LCA = %d\n", ans);

    return 0;
}

#endif

///////
int LCA( Tree T, int u, int v ) {
    int a, b;
    if (u>v) {
        a = v; b = u;
    } else { // u < v
        a = u; b = v;
    }
    // a < b
    int fa = 0;
    Tree Ta = T;
    while (Ta) {
        if (a > Ta->Key) Ta = Ta->Right;
        else if (a < Ta->Key) Ta = Ta->Left;
        else {
            fa = 1;
            break;
        }
    }
    int fb = 0;
    Tree Tb = T;
    while (Tb) {
        if (b > Tb->Key) Tb = Tb->Right;
        else if (b < Tb->Key) Tb = Tb->Left;
        else {
            fb = 1;
            break;
        }
    }

    if (!(fa&&fb)) return ERROR;

    while (1) {
        if (T->Key > a && T->Key > b) T = T->Left;
        else if (T->Key < a && T->Key < b) T = T->Right;
        else return T->Key;
    }
}
```

评测结果：答案正确（8 分）

测试点	结果	得分	耗时	内存
0	答案正确	1	4 ms	356 KB
1	答案正确	1	3 ms	256 KB
2	答案正确	1	3 ms	256 KB
3	答案正确	1	2 ms	256 KB
4	答案正确	1	2 ms	256 KB
5	答案正确	1	4 ms	256 KB

测试点	结果	得分	耗时	内存
6	答案正确	1	3 ms	368 KB
7	答案正确	1	2 ms	256 KB

```
a.c: In function ‘BuildTree’:
a.c:35:6: warning: ignoring return value of ‘scanf’, declared with attribute warn_unused_result [-Wunused-result]
      scanf("%d", &n);
      ^~~~~~~~~~~~~~
a.c:37:10: warning: ignoring return value of ‘scanf’, declared with attribute warn_unused_result [-Wunused-result]
          scanf("%d", &x);
          ^~~~~~~~~~~~~~
a.c: In function ‘main’:
a.c:51:5: warning: ignoring return value of ‘scanf’, declared with attribute warn_unused_result [-Wunused-result]
     scanf("%d %d", &u, &v);
     ^~~~~~~~~~~~~~~~~~~~~
```